



HackHere Community AIVENTRA

AI-Powered Forensic Triage & Postmortem Intelligence System

Problem Statement

Investigative agencies and forensic departments often face challenges in processing large amounts of forensic and digital evidence within a limited time. Manual analysis of autopsy reports, environmental conditions, and digital traces such as CCTV logs or mobile metadata can delay investigations and increase the possibility of oversight. In suspicious or complex cases, early-stage evaluation becomes critical for identifying important leads and prioritizing investigations. There is a growing need for an intelligent support system that can assist investigators by organizing, analyzing, and correlating forensic and digital evidence efficiently.

Core Innovations

The proposed system integrates Artificial Intelligence, Natural Language Processing (NLP), and data correlation techniques to support forensic investigations.

Key Features:

- **AI-Based Autopsy Report Analysis:** Extracts important forensic details such as probable cause of death, injury patterns, and medical observations from unstructured reports.
- **Time-of-Death Estimation:** Uses environmental conditions and postmortem indicators such as body temperature, rigor mortis, and livor mortis to generate approximate estimations.
- **Digital Evidence Correlation:** Connects and analyzes data from CCTV logs, timestamps, mobile metadata, and geolocation records to identify possible patterns and timelines.
- **Case Risk Scoring & Anomaly Detection:** Assists investigators by identifying suspicious patterns and generating structured case insights for faster triage.
- **Interactive Investigation Dashboard:** Provides summarized reports, evidence timelines, and visual insights for easier analysis and decision-making.

Ethical & Legal Considerations

The system is designed strictly as an investigative assistance platform and not as a replacement for forensic experts, medical professionals, or legal authorities. All outputs generated by the system are intended to support human decision-making rather than automate legal conclusions.

Data privacy, secure storage, and controlled access mechanisms will be considered during implementation. Explainable AI methods will also be incorporated to improve transparency and ensure responsible usage of the system in sensitive investigative environments.

Future Scope

Future improvements may include integration with larger forensic and crime databases, multilingual forensic report analysis, and real-time IoT sensor integration for environmental monitoring. The system can also be extended with federated learning approaches to enable secure collaboration between investigative agencies while maintaining data privacy.

Additional enhancements such as predictive investigation support, advanced behavioral analysis, and cross-case pattern recognition can further improve the effectiveness of forensic investigations.

Conclusion

The AI-Powered Forensic Triage & Postmortem Intelligence System aims to improve the efficiency and accuracy of forensic investigations through the use of AI-driven analysis and digital evidence correlation. By assisting investigators in organizing evidence, estimating timelines, and identifying patterns, the system can support faster and more informed investigative processes. This approach has the potential to reduce case delays, improve evidence handling, and contribute to stronger and more efficient justice systems.